Project Underdog: Can Tactics Make Quality Matter Less?

A pre-registered search for an underdog's tactical edge: four nulls, one small market effect, and one flat channel

Autonomous research run, 11 October 2026. Pre-registration committed before any outcome data were inspected (git `d94684e`). All deviations are logged in `DEVIATIONS.md`.

Abstract

We asked whether a weaker football team, such as Brentford against Manchester City, can systematically change the *structure* of a match so that player quality matters less, and what that would be worth in league points and title probability. We framed tactics as moves in a two-dimensional space: match **intensity** (total chances) and the underdog's **share** of chances. We then solved the in-match decision problem exactly by dynamic programming over (minute, goal difference). The empirical inputs were StatsBomb event data for 1,551 big-five league matches (2015/16) and Wyscout event data for 1,826 matches (2017/18), plus closing betting odds for 45,825 matches (2000–2026).

Five hypotheses were pre-registered. **Four did not survive.** (P1) A team's style in one stretch of a match does not predict its chances in the next stretch beyond strength and score: out-of-sample Δdeviance −0.6, 95% CI [−3.1, +2.0]. (P2) The optimal in-match style policy produced no reliable gain once evaluated honestly. Cross-fitted gain was +0.015 points/match, CI [−0.04, +0.08], against a pre-registered threshold of 0.05. The pre-registered criterion as literally worded *was* met, but only through the optimiser's curse. (P3) Habitual style did not predict results beyond the market. (P5) A "win-hungry" risk dial did not raise title probability, even for a contender. **One survived, modestly.** (P4) Over 36,374 underdog matches, underdogs that habitually play high-shot games underperform the closing market by 0.0069 points per extra shot per match (CI [−0.012, −0.002]). The effect is consistent across all five leagues but fades after 2015, and it does not follow the pattern pure variance theory predicts.

A post-hoc analysis found the most robust structural fact in the study, and it replicated across a second season and data provider. **Set pieces are the "flattest" source of chances.** Their quality elasticity is 0.41/0.28 against 1.09/0.89 for open play, and direct free kicks are near zero (0.12 in both). In mismatches, underdogs collect 37–38% of set-piece xG but only 30–32% of build-up xG. Set-piece efficiency is the only team skill we found that is both persistent and uncorrelated with team strength, but its persistence is modest (reliability 0.14–0.41).

Applied to Brentford's 2025/26 strength, every evidence-backed lever combined is worth **+1.8 to +4.5 points a season**. That lifts their top-four probability from 3% to 5–9%. Title probability stays at or below 0.15%. The honest answer to the project's question: there is a structure in which quality matters less (dead balls), but no tactic we could identify turns a mid-table squad into a title contender. The one underdog champion in our data, Leicester 2015/16, beat the market by 28 points, and 18 of them sat *above* their own chance quality.

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1. The question

The brief was to treat football as a sequential decision problem and look for a tactical policy, not a prediction model, that lets a weaker squad consistently outperform stronger ones. “Rediscovering” set pieces and direct play would not count; the goal was something materially better, with credible counterfactual validation.

Two traps were clear from the start, and the design is built around them.

1. **Historical data show what teams did, not what would have happened otherwise.** Style and outcomes are entangled. When the opponent attacks, our defensive actions are deep *because* they are attacking.
2. **An optimiser will exploit its own simulator.** A policy tuned on a fitted model always looks good when it is scored on the same model.

So we pre-registered the hypotheses, kept the 2015/16 Premier League out of model fitting, and evaluated every optimised policy on a fit it had not seen.

2. Data

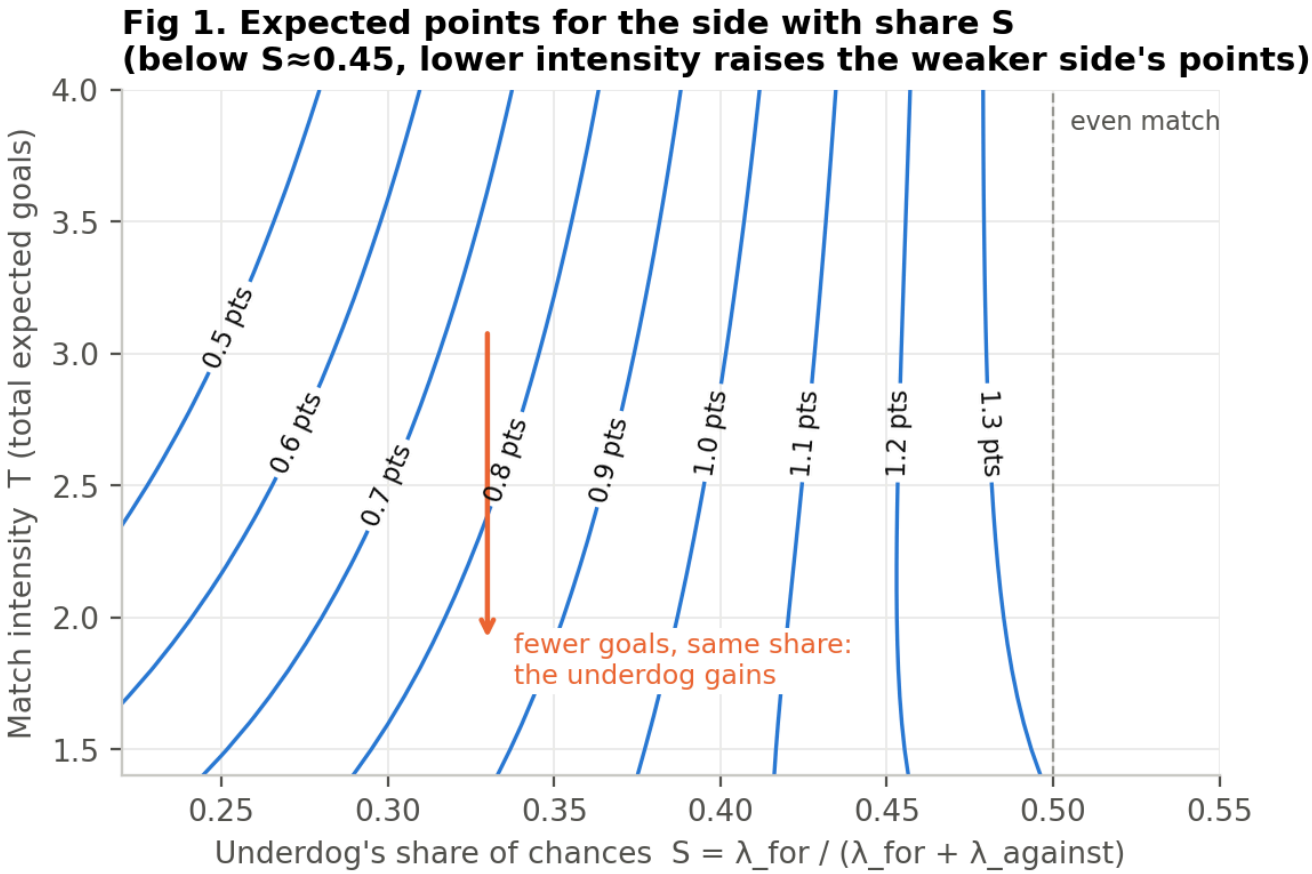
Source	Coverage	Used for
StatsBomb Open Data	1,551 matches, 2015/16: full EPL, La Liga, Serie A, Ligue 1, plus 34 Bundesliga. 5.4M events with xG	style, segments, coaching decisions, chance channels
Wyscout public dataset (Pappalardo et al., 2019)	1,826 matches, 2017/18, big five	out-of-season, out-of-provider replication
football-data.co.uk	45,825 big-five matches 2000/01–2025/26, closing odds + shots	market strength, P4, 2025/26 season simulation

Market strength. For every match we de-vig the closing 1X2 odds (Pinnacle closing where available) and invert them, together with the over/under 2.5 price when available, into Poisson goal rates (λ_home, λ_away) with a Dixon–Coles correction. This is the strength baseline throughout. It is demanding: anything the market already knows is not an edge.

Segments. Each match is cut into stretches of constant game state. Cuts fall at 15-minute marks, every goal and every red card, giving 27,570 team-segments. For each one we record xG for and against, and four style descriptors: defensive-action height, pressures per opponent pass, long-pass share and pass share.

3. Framework: the underdog’s exchange rate

Write a team’s chance production as λ_for and λ_against. Any tactic moves the match along two axes: **intensity** $T = \lambda_for + \lambda_against$, and **share** $S = \lambda_for / T$. Figure 1 shows expected points for the side with share S. Below $S \approx 0.45$ the iso-point lines tilt, so at a fixed share **fewer total goals raise the weaker side’s points.** More draws, and more noise relative to quality.

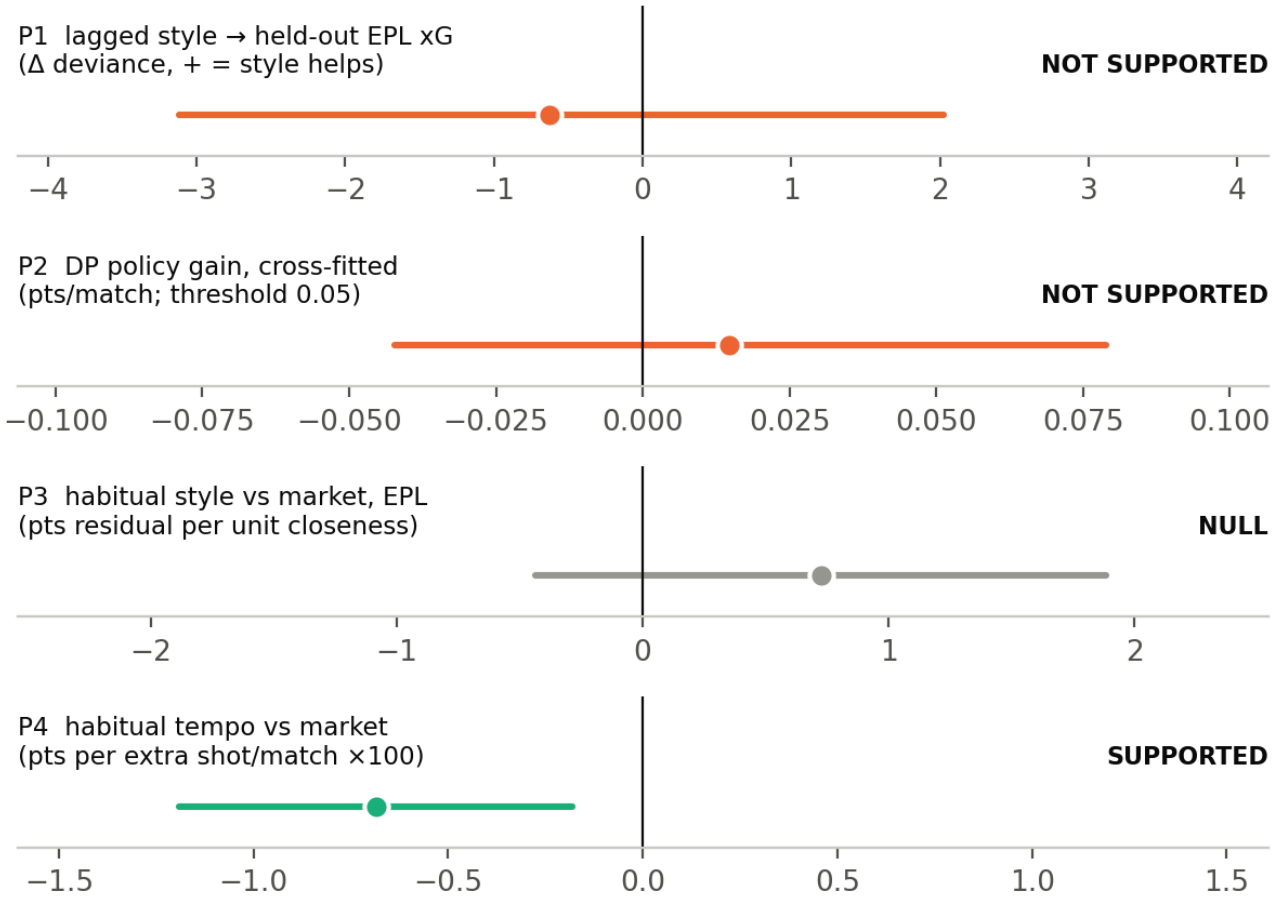


The exchange rate is concrete. For a 30%-share underdog in a 2.7-goal match, cutting total intensity by one goal is worth +0.10 points/match, the same as gaining 3.4 percentage points of share. For a 40%-share side the same cut is worth only +0.045. This is the theoretical case for “shrinking the game”. The empirical question is whether any lever moves T without giving back more than that in S.

The in-match decision problem is solved exactly. The state is (minute, goal difference), the actions are style modes with per-minute scoring rates, and the terminal utility is league points (3/1/0) or any other utility. Backward induction gives the optimal policy, and a forward pass gives the outcome distribution (`src/dp.py`).

4. Pre-registered results

Fig 2. Pre-registered tests (point estimate, 95% CI). P5 shown in Table 3



	Hypothesis (registered direction)	Result	Verdict
P1	Lagged style improves held-out EPL xG prediction	Δdev −0.62, CI [−3.12, +2.02]	**Not supported**
P2	DP policy gains ≥ 0.05 pts/match for a heavy underdog, robust to refitting	cross-fitted +0.015, CI [−0.042, +0.079]; positive in 70% of refits	**Not supported**
P3	Habitual style near the P2 prescription beats the market (EPL)	+0.72, CI [−0.44, +1.88]; n = 330	**Null**
P4	Habitual tempo negatively predicts underdog residual vs market	−0.0069 pts/shot, CI [−0.0119, −0.0018], p = 0.008; n = 36,374	**Supported (modest)**
P5	Title-maximising policy is more variance-seeking than points-maximising	title probability flat in κ (Table 3)	**Not supported**

P1. With style lagged by one segment (the primary specification, fixed in DEVIATIONS #1 before results were seen), style adds nothing out of sample. The contemporaneous model "improves" deviance by 7.1% (CI [+5.5%, +8.9%]), but its signs show it is mechanical. The opponent's defensive height is the strongest "predictor" of our xG (−0.28 log-points per SD) because deep opponents are a *consequence* of our attacking. A naive analysis would have reported that result as a discovery.

P2. The registered frontier, the lagged-style GLM with 27 style actions within ±1 SD of observed support, recommends pressing high and going long. Evaluated three ways:

Evaluation	Gain (pts/match)
Naive: optimise and evaluate on the same fit	+0.027
Literal registered criterion: refit on 60 bootstrap resamples, re-optimise	mean +0.051, **100%** of resamples > 0
Cross-fit: optimise on resample *b*, evaluate on independent resample *b'*	**+0.015, CI [−0.042, +0.079], 70% > 0**

The literal criterion is met, but it is trivially satisfiable: an optimiser that picks the best of 27 noisy actions always finds a positive gain on its own fit. We regard this as a flaw in our pre-registration wording and report the cross-fitted number as the real result. This is the "simulator exploitation" failure mode the brief warned about, measured. In-sample gains overstate the honest gain by about 3×.

P4. Underdogs whose previous 10 league matches featured more total shots underperform their closing-market expectation. The sign is negative in every league (EPL −0.0075, Bundesliga −0.0085, Ligue 1 −0.0084, Serie A −0.0076, La Liga −0.0031). It survives controlling for the team's own shot share (−0.0065, p = 0.012). It is strongest in 2005–14 (−0.0113, p = 0.003) and indistinguishable from zero since 2015 (−0.0033, p = 0.34), consistent with a market that has learned. It is null for heavy underdogs (p_win < 0.25: −0.0025, CI [−0.0088, +0.0038]). At one SD of habitual tempo (2.3 shots) the effect is ~0.016 points/match, about 0.6 points/season.

Is P4 the variance mechanism? Probably not. If the market ignored variance entirely, the residual would equal the Poisson-theory effect of tempo. Pooled, theory and data agree (−0.0052 vs −0.0069 per shot), but by strength band they diverge (`results/theory_vs_p4.json`):

Underdog p_win	n	Empirical	Variance theory
< 0.20	14,324	−0.0018 (0.0036)	−0.0104
0.20–0.30	14,442	−0.0060 (0.0042)	−0.0035
0.30–0.45	7,608	−0.0177 (0.0062)	+0.0014

Theory says heavy underdogs should gain most from low tempo; the mispricing is instead concentrated in *mild* underdogs, where theory predicts nothing. The market appears to price the variance effect for long shots and to overrate high-tempo "chaos" teams that are only slightly weaker. That is a market-efficiency finding, not a tactical one.

5. Post-hoc analyses (hypothesis-generating)

5.1 Coaching decisions barely move chance rates

After P1 failed, we turned to the levers coaches actually pull in-game. There were 6,789 substitution and formation-change decisions, each scored by the change in the team's positional attacking load (`src/tactical_changes.py`). Post-decision xG over the next ≤15 minutes (stopping at the next goal) was compared with a strength + state baseline, controlling for pre-decision momentum and the opponent's simultaneous changes.

Decision (vs like-for-like)	Own xG	xG conceded
Attacking (n = 539)	×1.04 (SE 0.06)	×1.03 (SE 0.06)

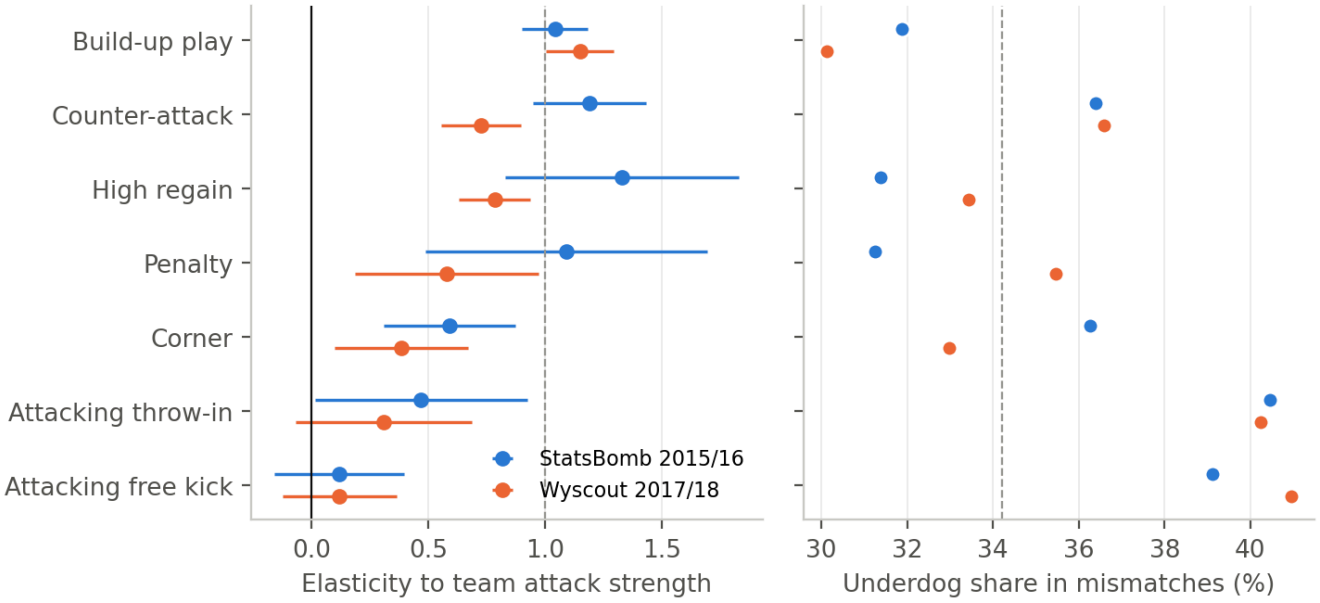
Decision (vs like-for-like)	Own xG	xG conceded
Defensive (n = 488)	×1.13 (SE 0.07)	×1.03 (SE 0.07)

Neither kind of change detectably moves either rate. The DP built on these effects shows a cross-fitted gain of +0.035, CI [−0.075, +0.185]. What does move rates is game state itself: trailing teams' xG rises steeply late in matches (late × trailing: +0.32 log-points per 90 min). Every team already responds to the score; deliberate changes add little on top that we can detect.

5.2 Where quality matters less: chance channels

We split every possession into channels by origin: build-up, counter-attack (regain → shot ≤ 15 s, or tagged), high regain (won in the final third), corner, attacking free kick, attacking throw-in and penalty. Restarts count as set pieces only if any shot comes within 20 s. Each channel's team-match xG was then regressed on market attacking strength.

Fig 3. Where quality matters less: set pieces are flat, build-up is steep



Channel	Elasticity: StatsBomb 15/16	Elasticity: Wyscout 17/18	Underdog share in mismatches (SB / WY)
Build-up	1.04 (0.07)	1.15 (0.07)	31.9% / 30.1%
Counter	1.19 (0.12)	0.73 (0.09)	36.4% / 36.6%
Corner	0.59 (0.14)	0.39 (0.15)	36.3% / 33.0%
Attacking throw-in	0.47 (0.23)	0.31 (0.19)	40.5% / 40.2%
Attacking free kick	0.12 (0.14)	0.12 (0.12)	39.1% / 41.0%
All open play	**1.09**	**0.89**	**33.0% / 33.1%**
All set pieces	**0.41**	**0.28**	**38.0% / 36.9%**

Elasticity 1 means a team with 10% more attacking strength creates 10% more xG in that channel. The Wyscout elasticities are uniformly lower because its xG is our location-only model (AUC 0.78 vs StatsBomb's 0.815), which compresses chance-quality differences. Compare ratios, not levels.

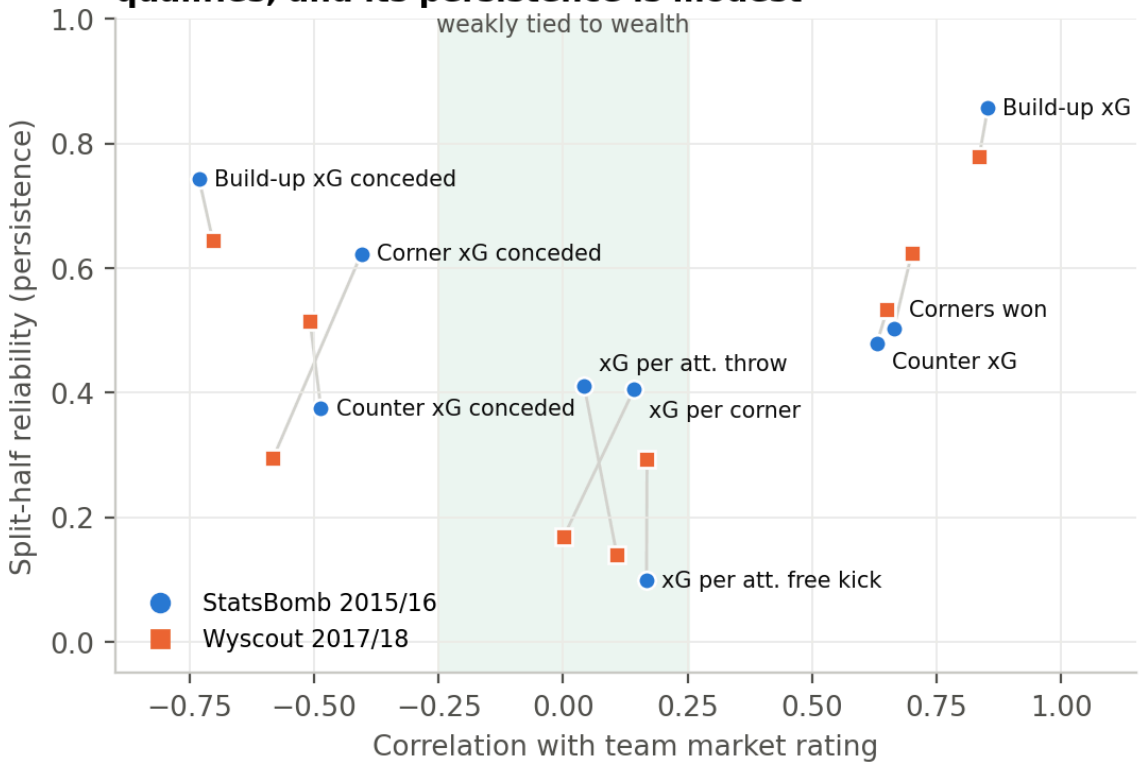
What replicates: **set pieces are roughly a third as quality-sensitive as open play**, in both datasets, with different xG models and two seasons apart. **Build-up play is where underdogs lose most** (30–32% share). Underdogs also keep a relatively large share of counter-attack xG (36–37%) in both datasets.

What does not replicate: the counter-attack *elasticity* (1.19 vs 0.73). We drew an early "counters are not an equaliser" conclusion from the first dataset and withdrew it.

5.3 The cheap-skill map

A useful underdog lever is a team skill that is **persistent** (a real ability, not noise) and **not bought** (uncorrelated with team strength). We computed split-half reliability (odd vs even matches, Spearman–Brown corrected) and the correlation with the team's market rating for each metric.

Fig 4. Cheap skills (persistent, not bought): only set-piece efficiency qualifies, and its persistence is modest



Most persistent metrics are wealth-bound: build-up xG for and against have reliability 0.65–0.86 and |r| = 0.70–0.85 with rating. Only set-piece *efficiency* sits in the cheap region. xG per corner has r = 0.14/0.00 with rating and xG per attacking throw r = 0.04/0.11. Their persistence is modest, though: 0.41 for both in StatsBomb, and 0.17/0.14 in Wyscout. Corner *defence* is more persistent (0.62/0.30) but is partly wealth-linked (r = −0.41/−0.58). This is Brentford's existing edge. The data say it is real but small, and noisy enough that a single season cannot confirm a team has it. It is also already priced: a team's habitual net set-piece xG does not predict its points against the closing market (StatsBomb +0.07 ± 0.22 per xG/match; Wyscout −0.13 ± 0.24; `results/setpiece_market.json`). A club can still *build* the skill cheaply, but it cannot hide it.

6. Brentford, 2025/26

We fitted attack/defence ratings to the 2025/26 Premier League closing-market rates and simulated 200,000 seasons. Non-Brentford matches use Dixon–Coles Poisson scorelines; Brentford matches use the DP outcome distribution under each intervention. Baseline Brentford: 50.2 expected points (actual: 53), 11th on average.

Interventions are restricted to observed support:

- **Set pieces to the 90th percentile** of *true* team skill (observed SD × √reliability). That is +0.058 xG/match for and −0.052 against with StatsBomb reliabilities, or +0.036/−0.036 with Wyscout's.
- **Tempo menu** {×0.91, ×1, ×1.09} on both teams' rates at a fixed share. ±9% is one SD of the between-team spread in habitual tempo (P4 data). The DP chooses by (minute, score). This is an **upper bound**: P1, P2 and §5.1 found no in-game lever that does this reliably.

Fig 6. Brentford: what the evidence-backed playbook is worth over 38 matches

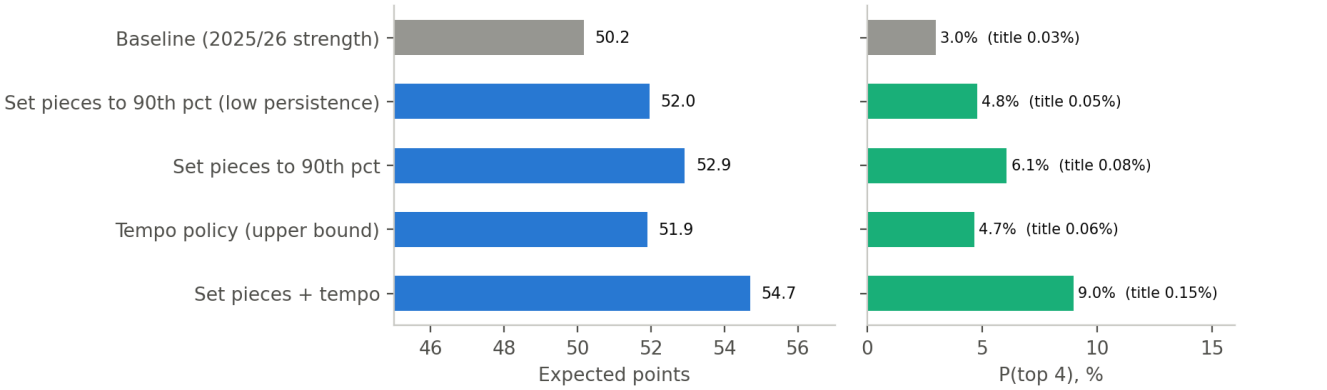


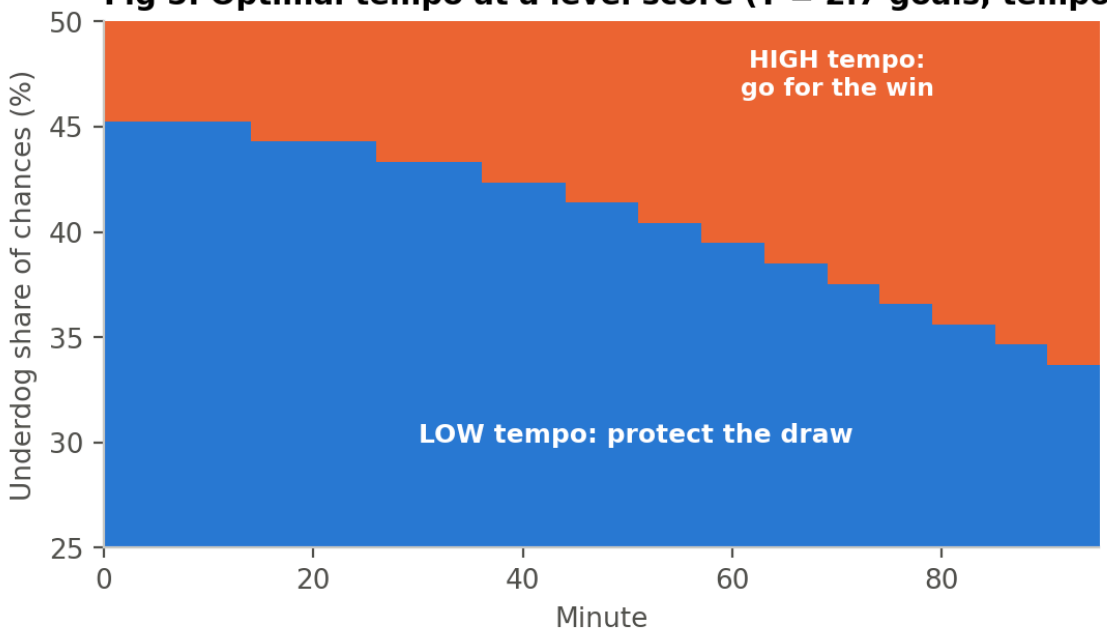
Table 3. Brentford season outcomes (200k seasons; risk-dial rows 300k, common random numbers).

Scenario	E[pts]	P(top 4)	P(title)
Baseline	50.2	3.0%	0.03%
Set pieces p90 (low persistence)	52.0	4.8%	0.05%
Set pieces p90	52.9	6.1%	0.08%
Tempo policy (upper bound)	51.9	4.7%	0.06%
Set pieces + tempo	54.7	9.0%	0.15%
… + risk dial κ = 1, 3, 8 (P5)	54.7	9.0%	0.15%
Reference: +10% attacking quality	*53.8*	*7.5%*	*0.11%*
Reference: +40% attacking quality	*65.6*	*48.6%*	*3.6%*

The full playbook is worth about the same as a 10–12% improvement in attacking quality. P5 was also tested at contender strength (+40% and +60% attack; title probability 5.7% and 25.9%). The risk dial $\kappa \in \{0, 1, 3, 8\}$, which values wins above draws, never raised title probability and slightly lowered it in both cases. Over a 38-match season, maximising expected points is title-optimal.

When to open up. Figure 5 reads the "drawing at 70 minutes" question straight off the value function. With a ±9% tempo menu and level scores, a side with under ~45% of the chances should play low-tempo from kick-off. The share at which it should switch to high tempo falls steadily through the match, to about 34% by full time.

Fig 5. Optimal tempo at a level score (T = 2.7 goals, tempo menu ±9%)



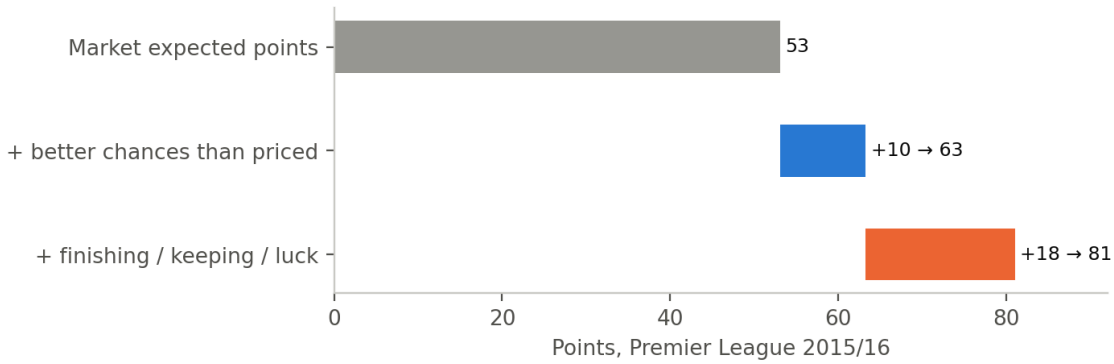
For Brentford vs Manchester City (2025/26 strengths), the expected points and policy are:

	Home (λ 1.13 vs 1.95)	Away (λ 0.93 vs 2.37)
Baseline E[pts]	0.872	0.566
DP tempo policy	0.910	0.606
Set pieces + DP tempo	0.975	0.659
Level at 70'	protect (low)	protect (low)
Level at 85'	**go (high)**	protect (low)
Trailing by 1	go (high) from kick-off	go (high) from the 45th minute

So the answer to the brief's question is: **at home, keep the draw until about the 85th minute, then go for it; away, keep the point.** All of this is conditional on the tempo lever existing.

7. Leicester 2015/16: the one underdog champion

Fig 7. Leicester 2015/16: two thirds of the miracle sat above their chances



Leicester's market-expected points were 53.1; they finished on 81. Their xG-implied expected points were 63.3. So roughly 10 points came from creating better chances than the market priced, and 18 from finishing, goalkeeping and fortune above their own chance quality. Their style did not shrink the game. They ranked **1st in tempo** (2.96 xG per match, both teams), **1st in counter-attack share** (24% of their xG), 18th in

possession, 4th in long-ball share. They also had by far the most penalty xG (10.2). They opened matches up, which is the opposite of the variance prescription. The rare underdog title in our data looks like underpriced talent (Vardy, Mahrez, Kanté) plus a large positive draw from the noise. We found no reusable structural trick in it.

8. What this means

1. **"Make quality matter less" has one robust answer: dead balls.** Set pieces are roughly a third as sensitive to team quality as open play. That held in two seasons, two providers and two xG models. Brentford already exploit this, as the brief anticipated. Our contribution is to quantify the ceiling: about 2–3 points a season for reaching the 90th percentile.
2. **Build-up play is where the gap is widest.** Underdogs keep only 30–32% of build-up xG in mismatches. A weaker team that insists on out-building stronger teams is fighting in the steepest channel.
3. **The in-game "AI coach" did not survive honest evaluation.** Neither continuous style knobs nor discrete coaching changes showed effects we could distinguish from zero once reverse causation and the optimiser's curse were removed. The tempo policy in §6 is what a coach *could* gain if a reliable ±9% intensity lever existed. With the event data available publicly, we could not show that it does.
4. **The market already knows most of it.** The only pre-registered effect that survived (P4) is small, concentrated in mild underdogs, inconsistent with pure variance theory, and has faded since 2015.
5. **For a title, tactics are not the binding constraint.** Every evidence-backed lever together moves Brentford from 0.03% to 0.15%. A title needs roughly +40% attacking quality *and* luck.

9. Limitations

- **Identification.** Every frontier is observational. Lagging, game-state control, like-for-like control groups and cross-fitting reduce but cannot remove confounding. A coach-run controlled trial (milestone 4 in the brief) is the only clean test.
- **Event data, not tracking.** Pressing structure, defensive line height and off-ball runs are only proxied by on-ball events. The levers we could not measure may be the ones that matter. Licensed tracking data (e.g. 2025/26 Premier League) is the obvious next step and could overturn P1/P2.
- **Quality measure.** Market odds include the effect of a team's habitual style, so "wealth correlation" uses a rating that is not purely about player quality. Transfermarkt squad values were not accessible to our pipeline.
- **Single seasons.** Cheap-skill reliabilities come from one season per dataset (81 and 98 team-seasons) and differ between datasets.
- **Season model.** Brentford's set-piece baseline is assumed median. If they are already above median (likely), the headroom is smaller than shown.
- **Pre-registration wording.** P2's literal criterion could be met by noise. We flag this rather than claim it.

10. Reproducibility

Everything runs from `underdog/src/` in this order: `fetch_statsbomb.py`, `odds.py`, `build_segments.py`, `p4_tempo_market.py`, `p4_robust.py`, `p1_heldout.py`, `tactical_changes.py`, `event_study.py`, `p2_policy.py`, `possessions.py`, `channels.py`, `p3_style_market.py`, `theory_vs_p4.py`, `xg_model.py`, `wyscout_replication.py`, `season_sim.py`, `season_sensitivity.py`, `p5_and_city.py`, `leicester.py`,

`setpiece_market.py`, `make_figures.py`. Every number in this paper is in `underdog/results/*.json`. The seed for all bootstraps is 20261011.

Data: StatsBomb Open Data (attribution required); Pappalardo et al. (2019), "A public data set of spatio-temporal match events in soccer competitions", Scientific Data 6:236; football-data.co.uk.